Does Remote Sensing Beat the Trail Map? Predicting Hiking Speed in Unseen Regions from Lidar Terrain and OpenStreetMap Tags

Brian Zhou [TODO: author list and order: ask Danni, Tianyi and Prof. Jacobs]
Department of Computer Science and Engineering, Washington University in St. Louis

Abstract—[TODO: 150–200 words, written last. Problem: travel time on foot; Tobler. Data: 17 regions, 290k walking segments from public OSM GPS traces. Test: leave one region out, calibrated predictive distributions (CRPS). Result: terrain from airborne-lidar elevation transfers to unseen regions (+2.1 to +2.8% CRPS skill over a refitted slope curve), map tags barely do (+0.4 to +0.9%), satellite embeddings help within regions but transfer less reliably; Tobler is 12.5% worse. A terrain-adjusted hiking function needs one DEM-derived input.]

Index Terms—hiking speed, lidar, OpenStreetMap, GPS trajectories, transfer, calibration

I. Introduction

[TODO: Paragraph 1, why travel time on foot matters: search and rescue, wildfire escape routes, trail apps, accessibility maps. GIS still uses Tobler's 1993 hiking function [1], $v = 6\exp(-3.5\,|s + 0.05|)$ km/h, and fixed multipliers for other modes.]

[TODO: Paragraph 2, what is known: crowdsourced GPS shows Tobler is too fast and too peaked [2], [3]; Wood et al. [4] combine OSM terrain classes with lidar; STRIDE [5] models walking speed from lidar slope, understory density [6] and roughness, calibrated on field walks; RasterNet [7] learns road speed from lidar and imagery.]

[TODO: Paragraph 3, the gap: no work tests whether terrain covariates transfer to regions a model has never seen, against the map tags used today, with calibrated uncertainty.]

[TODO: Paragraph 4, contributions (three bullets): (1) a 17-region dataset of walking traversals of 50 m trail segments from public OSM traces; (2) an evaluation protocol (leave one region out, CRPS, coverage) with published baselines; (3) the finding that elevation-derived terrain transfers and map tags do not, plus a one-input terrain-adjusted hiking function.]

II. Data

[TODO: Sources: OSM GPS traces [8] from the trackpoints API (5,000 points a page), the original files of named uploads, and a lab crawl of 1.5M original files (2016–2022). 17 regions in the US and Canada (Fig. 1); 16 from full downloads, the Bay Area from the crawl only.]

[TODO: Pipeline in one paragraph: duplicates and group hikes merged into outings; stops removed (60 s / 10 m window); hidden-Markov map matching to OSM

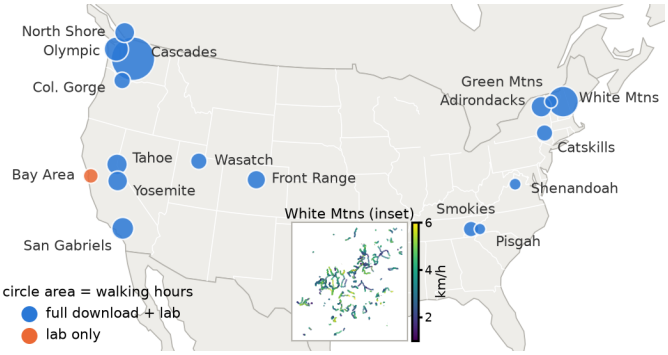


Fig. 1. The 17 regions. Circle area is walking hours (1,994 outings, 4,702 h in total). Blue: full OSM download plus the lab crawl; orange: lab crawl only. Inset: the White Mountains' 34,535 walking segments colored by speed.

trails [9] (97% of points matched); 50 m segments along the matched path; slope from USGS 3DEP [10]; travel mode anchored on foot-only trails, checked against 327 traces whose own app named the activity (5% of "walkers" labeled bike or ski).]

[TODO: Numbers: 2,876 hiking traces, 1,994 walking outings, 4,702 moving hours; 290,640 walking segments (14,532 km). 1 m lidar under at least 83% of walked trail in 13 regions.]

[TODO: Limits in one sentence each: uploaders are enthusiasts; busy API tiles cannot reach their oldest uploads; the Bay Area lacks anonymous traces.]

III. Methods

[TODO: Target: the distribution of $\ln v$ for a walking trail segment.]

[TODO: Covariates. (a) Slope. (b) OSM tags: sac_scale, trail_visibility, surface, width, smoothness, mtb_scale, highway, access. (c) Terrain from elevation, read every 1 m along the segment's matched path: spread and maximum of 5 m pitches, relief along the line of travel, side slope (5 m grids; 1 m lidar adds STRIDE's roughness). (d) NLCD 30 m canopy and land cover. (e) AlphaEarth satellite embeddings [11]. Control: "hidden climbing", the extra time a fixed Tobler curve predicts when the segment is walked as 5 m pieces.]

[TODO: Models: published Tobler, Campbell (median) [3] and STRIDE [5]; a refitted rounded asymmetric

TABLE I
CRPS skill (%) vs the refitted slope curve. Additive / trees; 95% intervals for unseen regions resample whole regions.

Model	New hikers	New 10 km	Unseen region
Tobler 1993	−12.8	−12.7	−12.5
Campbell 2022	−0.9	−0.9	−0.5
STRIDE 2024 (refit)	−17.2 (+0.2)	−17.3 (+0.2)	−16.8 (+0.2)
OSM tags	+1.3 / +1.5	+1.1 / +1.5	+0.9 / +0.4
30 m vegetation	+0.0 / +0.1	−0.0 / −0.1	−0.1 / −0.8
Embeddings	+1.4 / +2.6	+1.4 / +1.9	+1.1 / +1.1
Terrain	+2.2 / +3.2	+2.1 / +2.8	**+2.1** / **+2.8**
Terrain + emb. + tags	+3.7 / +5.1	+3.8 / +5.1	+3.0 / +3.9
Eq. (1), one input	+1.0	+1.0	+1.0

slope curve (the baseline); additive ridge offsets on binned features; gradient boosted trees.]

[TODO: Predictive distribution: prediction plus residual quantiles from held out training outings, so calibration has to transfer too.]

[TODO: Splits (never by segment): 5 folds of outings; 5 folds of 10 km blocks; leave one region out (17 folds). Score: CRPS [12] as skill vs the slope curve, 95% intervals resampling whole regions; coverage of 90% intervals; moving-time error per hike. References: the hiker's pace from their first km; each region's own offset.]

$$\ln v = a_0 + a_1 z - (k_0 + k_1 z)\,|s + s_0| \qquad (1)$$

with z the SD of 5 m pitches along the path divided by 0.10. [TODO: Introduce Eq. (1), the terrain-adjusted hiking function: Tobler's form, with peak speed and slope sensitivity depending on one DEM-derived number.]

IV. Results

[TODO: R1: Tobler fails everywhere (Table I): −12.5%, peak 4.31 km/h at −4.5% grade, not 6. Campbell's published curve ties our refit on average but is region-dependent (−10% in the Adirondacks).]

[TODO: R2: tags do not transfer: +1.3% within regions, +0.9% / +0.4% in unseen ones.]

[TODO: R3: terrain transfers: +2.1% / +2.8% in unseen regions, beating tags in 16 of 17 (Fig. 3); terrain minus tags +0.7 to +2.2, above zero in every bootstrap draw. About 60% of the gain is uneven grade inside the segment. Not a region proxy (adds ∼1.9 points over region offsets), not a hiker proxy. 1 m lidar ≈ 5 m; STRIDE's roughness adds nothing on trails; 30 m vegetation adds nothing; embeddings help within regions but transfer less reliably.]

[TODO: R4: the hiker dominates whole-hike time (Fig. 4): median error 13.6% (slope) → 13.2% (terrain) → 9.5% (plus the hiker's first km).]

V. Discussion and Conclusion

[TODO: Why tags do not transfer: mapping communities tag differently; the same sac_scale means different things in different places. Elevation data measures the same physical thing everywhere.]

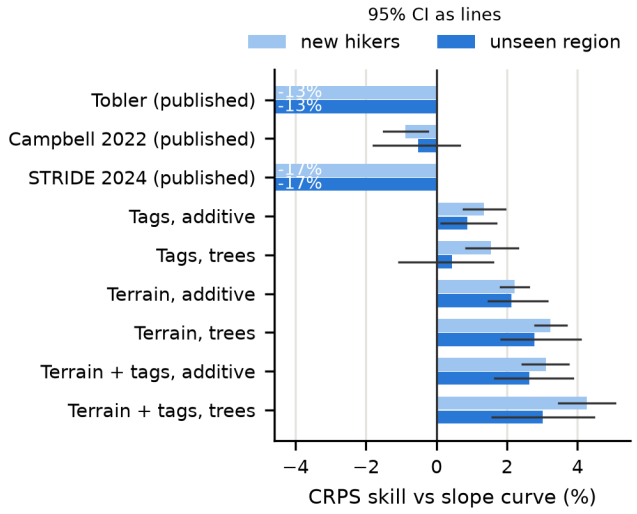


Fig. 2. CRPS skill vs the refitted slope curve for new hikers in known regions (light) and in regions the model never saw (dark), with 95% intervals.

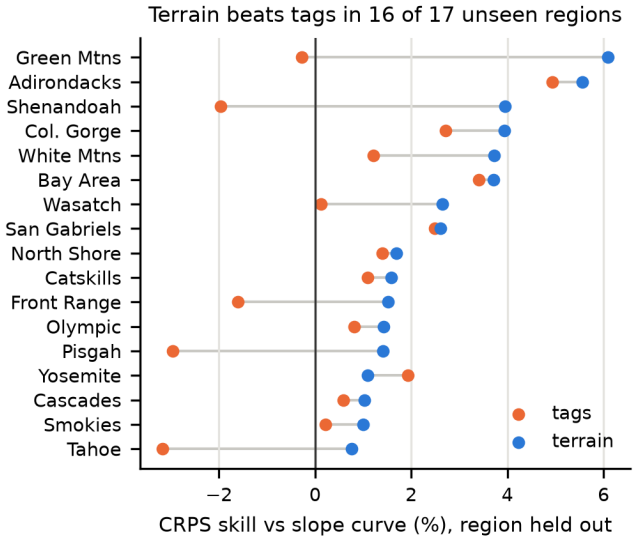


Fig. 3. Per held-out region: terrain vs OSM tags (additive models).

[TODO: What terrain captures: uneven grade within the segment (rock steps, short pitches) that one average slope hides, plus difficulty beyond grade. STRIDE's micro-roughness, calibrated off-trail, over-penalizes trails.]

[TODO: Practical output: Eq. (1) runs anywhere a DEM exists (Fig. 5); travel-mode multipliers fail for bikes (ratio ∼4× on descents, ∼1.6× on climbs).]

[TODO: Limits: OSM uploaders are not a random sample of hikers; per-segment effects are a few percent because GPS noise and the hiker dominate; no weather or season covariates yet; lidar understory density not tested (point-cloud cost).]

[TODO: Conclusion: three sentences. Data release (seg-

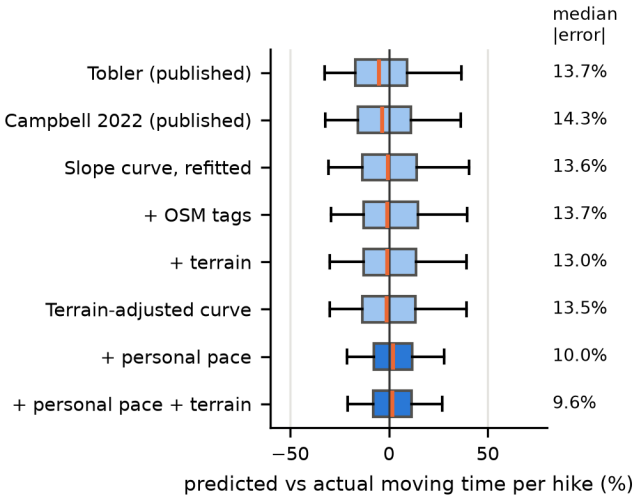


Fig. 4. Error in predicted moving time per hike, 1,354 hikes in held-out regions.

ments without user names) if the mentors agree.]

References

[1] W. Tobler, "Three presentations on geographical analysis and modeling," National Center for Geographic Information and Analysis (NCGIA), Tech. Rep. 93-1, 1993.

[2] M. J. Campbell, P. E. Dennison, B. W. Butler, and W. G. Page, "Using crowdsourced fitness tracker data to model the relationship between slope and travel rates," Applied Geography, vol. 106, pp. 93–107, 2019.

[3] M. J. Campbell, P. E. Dennison, and M. P. Thompson, "Predicting the variability in pedestrian travel rates and times using crowdsourced GPS data," Computers, Environment and Urban Systems, vol. 97, 2022.

[4] A. Wood et al., "Todo: check title and authors," PLOS ONE, 2023, TODO: verify every field before citing.

[5] M. J. Campbell, S. L. Cutler, and P. E. Dennison, "A singular, broadly-applicable model for estimating on- and off-path walking travel rates using airborne lidar data," Scientific Reports, vol. 14, p. 21393, 2024.

[6] M. J. Campbell et al., "Quantifying understory vegetation density using small-footprint airborne lidar," Remote Sensing of Environment, vol. 215, pp. 330–342, 2018.

[7] A. Hadzic, H. Blanton, W. Song, M. Chen, S. Workman, and N. Jacobs, "RasterNet: Modeling free-flow speed using LiDAR and overhead imagery," in IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (EARTHVISION), 2020.

[8] OpenStreetMap contributors, "OpenStreetMap," 2026, GPS traces and map data, ODbL; https://www.openstreetmap.org.

[9] P. Newson and J. Krumm, "Hidden Markov map matching through noise and sparseness," in ACM SIGSPATIAL International Conference on Advances in Geographic Information Systems, 2009.

[10] L. J. Sugarbaker, D. F. Eldridge, A. L. Jason, V. Lukas, D. L. Saghy, J. M. Stoker, and D. R. Thunen, "Status of the 3D elevation program, 2015," U.S. Geological Survey, Open-File Report 2016-1196, 2017.

[11] Google DeepMind, "AlphaEarth Foundations satellite embedding dataset," 2025, CC-BY 4.0; accessed via the Source Cooperative mirror tge-labs/aef. TODO: cite the AlphaEarth Foundations paper.

[12] T. Gneiting and A. E. Raftery, "Strictly proper scoring rules, prediction, and estimation," Journal of the American Statistical Association, vol. 102, no. 477, pp. 359–378, 2007.

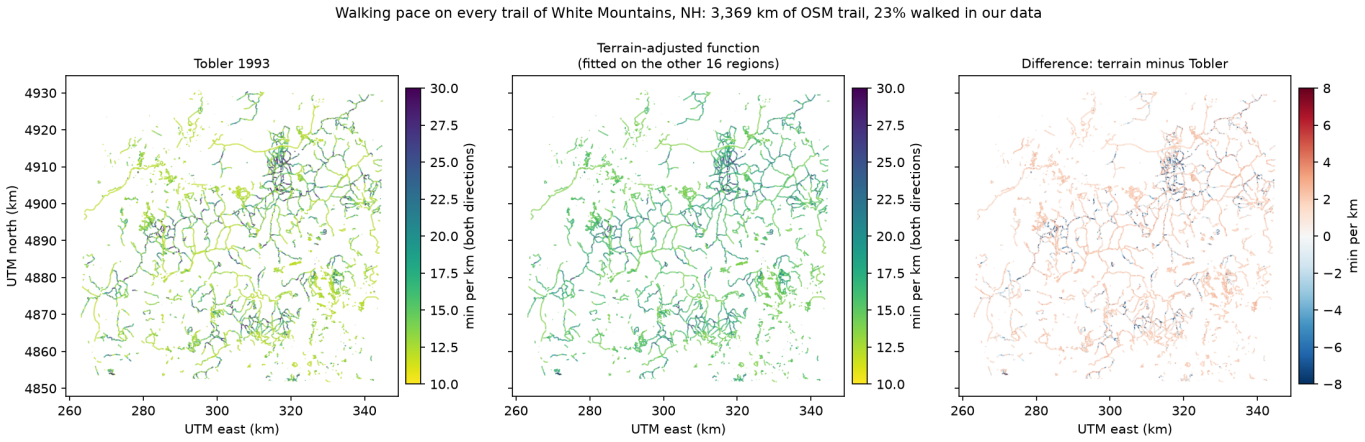


Fig. 5. Walking pace on every OSM trail of the White Mountains (3,369 km, 77% never walked in our data) from Eq. (1) fitted on the other 16 regions, vs Tobler. [TODO: Keep only if space allows; a two-panel crop may fit one column.]